

Biology — Chapter 4: Cell Cycle

Federal Board of Intermediate and Secondary Education, Islamabad | Annual Examination 2026 | SSC-I | Version 1

Candidate: SeemabExam: exam-03-ch4Attempt date: 2026-06-08Total Marks: 65Time allowed: 2h 30m

62.5 / 65

Section A: 12/12

Section B: 31.5/33

Section C: 19/20

96.2%

Rank: A+

A+

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Topic	Correct Answer	Seemab's Answer	Result
1	Two main phases of the cell cycle	C — Interphase and mitotic phase	C	✓
2	DNA replication phase of interphase	B — S phase	B	✓
3	Where spindle fibres attach to chromosomes	C — Centromere	C	✓
4	Longest phase of mitosis	D — Prophase	D	✓
5	Where chromosomes align in metaphase	B — Equator of spindles	B	✓
6	Plant cell spindle lacks	C — Centrioles	C	✓
7	Bivalent composition	B — 4 chromatids, 2 centromeres	B	✓
8	When crossing over occurs	C — Prophase I	C	✓
9	Daughter cells produced by meiosis	D — 4 haploid cells	D	✓
10	What halves chromosome number in meiosis	B — Separation of homologous chromosomes in meiosis I	B	✓
11	Gametes of fruit fly (2n=8)	B — 4 chromosomes	B	✓
12	Sea star arm regeneration — cell division type	C — Mitosis	C	✓
MCQ Score				12 / 12

SECTION B — Short Questions (11 × 3 = 33 Marks)

Q.2(i) — Main question

Cell cycle definition and two main phases

3 / 3

Definition of cell cycle given correctly: sequence of events involving growth of a newly formed cell, replication of its genome, and division into two daughter cells. +1

Interphase defined as the period between two consecutive divisions, divided into G1, S, G2 phases. +1

Mitotic phase (M phase) correctly described as the phase where cells divide into daughter cells. +1

Full marks.

3 / 3

Q.2(ii) — Main question

Three sub-phases of interphase (G1, S, G2)

3 / 3

G1 phase: cell grows in size, increases number of organelles, forms needed proteins and other substances. +1

S phase (Synthesis): DNA replication — cell replicates entire genetic material to form two copies; chromosomes get two chromatids; each attached to one centromere, called sister chromatids. +1

G2 phase: cell prepares for mitotic phase; forms spindle proteins; accumulates energy to complete cell division; more mitochondria are formed. +1

All three sub-phases clearly described with key events. Full marks.

3 / 3

Q.2(iii) — OR variant

Metaphase and anaphase of mitosis; role of spindle fibres in anaphase

3 / 3

Metaphase: chromosomes aligned to equator of spindles; two spindle fibres, one from each pole, attached to the centromere of chromosomes. +1

Anaphase: spindle fibres pull the centromeres (which split); separated chromatids pulled from each other; individual chromatids pulled to their respective poles. +1

Role of spindle fibres in anaphase: spindle fibres exert a pulling force on the centromeres, ensuring each pole receives an identical set of chromosomes. +1

All three mark points covered. Full marks.

3 / 3

Q.2(iv) — Main question

Telophase of mitosis

3 / 3

Chromatids reach their poles and uncoil/lengthen to reform chromatin fibres. **+1**

Spindle apparatus disintegrates; nuclear membrane reforms around chromatin at each pole. **+1**

Nucleoli reappear in both nuclei. **+1**

All three events stated precisely. Full marks.

3 / 3**Q.2(v) — OR variant**

Regeneration: definition and three organisms with body parts

3 / 3

Regeneration defined as the ability of some animals to regrow a whole body part if accidentally removed, with new cells produced by mitosis. **+1**

Three organisms listed: (1) Sea star — regenerates arm; (2) Earthworm — regenerates head; (3) Salamander — regenerates limb. **+1 each for three valid organisms with body parts = 2 marks**

Definition and all three organisms with correct body parts. Full marks.

3 / 3**Q.2(vi) — Main question**

Cancer, benign tumour, malignant tumour, metastasis

3 / 3

Cancer correctly defined as a disease of the cell cycle in which cells lack proper functioning of the cell cycle control system and divide excessively, resulting in an abnormal mass called a tumour. **+1**

Benign tumour: abnormal mass of normal cells that always remains at its original site. Malignant tumour: cancerous; capable of spreading over neighbouring tissues and to distant parts of the body. **+1**

Metastasis: spread of cancer cells beyond their original site. **+1**

All four terms defined correctly. Full marks.

3 / 3**Q.2(vii) — Main question**

Synapsis, bivalent, crossing over and chiasma

3 / 3

Synapsis: pairing of homologous chromosomes during prophase I, with centromeres in the same position. **+1**

Bivalent: each synapsed pair is called a bivalent. Non-sister chromatids join each other at points called chiasmata. **+1**

Crossing over: exchange of genetic material between non-sister chromatids at a chiasma, resulting in genetic recombination. **+1**

All terms defined precisely and correctly. Full marks.

3 / 3**Q.2(viii) — OR variant**

Telophase I and Cytokinesis I; type of cells produced

2.5 / 3

Telophase I: chromosomes uncoil at their poles; nuclear envelope reforms around the haploid set at each pole. **+1**

Cytokinesis I: occurs by cleavage in animal cells; by forming a cell wall in plant cells. Each daughter cell is haploid with chromosomes composed of two chromatids. **+1**

Result: correctly states two haploid daughter cells are formed and they enter Interphase II. **However, the specific point that chromosomes are NOT genetically identical due to crossing over (they must be separated in meiosis II) was stated in the answer — this scores the third mark partially.** The answer captures the essential content but the phrasing of "not genetically identical" without explicitly stating that this is WHY meiosis II is needed is slightly incomplete. **-0.5**

Telophase I + Cytokinesis I events present; result of two haploid cells stated. Minor incompleteness on the "entering interphase II" context.

2.5 / 3**Q.2(ix) — Main question**

No DNA replication in interphase II; meiosis II overview

3 / 3

Interphase II: varies in length but has no further DNA replication — chromosomes are already duplicated from S phase; replication would incorrectly double the genetic material again. **+1**

Meiosis II resembles mitosis; divided into prophase II, metaphase II, anaphase II, telophase II, during which sister chromatids are separated. **+1**

After cytokinesis II: four haploid daughter cells produced, all with new genetic combinations because of crossing over in meiosis I. **+1**

All three mark points covered. Full marks.

3 / 3**Q.2(x) — Main question**

Meiosis and maintenance of chromosome number (human example)

3 / 3

Correct explanation: humans have 46 chromosomes in somatic cells. Haploid gametes (egg and sperm) formed by meiosis have 23 chromosomes. **+1**

At fertilisation, sperm (23) fuses with egg (23) = zygote with 46 chromosomes, restoring the original number. **+1**

Meiosis therefore maintains the chromosome number across sexual reproduction. **+1**

Full human example with correct chromosome counts. Full marks.

3 / 3**Q.2(xi) — Main question**

Three differences between mitosis and meiosis

3 / 3

Clean two-column tabular comparison with correct three features:

(1) Location: somatic cells vs germ line cells. **+0.5 + 0.5 = 1**

(2) Daughter cells: 2 diploid daughter cells vs 4 haploid daughter cells. **+0.5 + 0.5 = 1**

(3) Chiasmata formation: no chiasmata vs chiasmata formed (crossing over). **+0.5 + 0.5 = 1**

Table format is clean and all entries are accurate.

Three differences correctly tabulated. Full marks.

3 / 3

Q.3(i) — Main question

Cell cycle in detail: G1, S, G2, G0 sub-phases and M phase

5 / 5

Definition of cell cycle: sequence of events involving growth, replication and division into two daughter cells. Two main phases named (interphase and mitotic phase). **+0.5**

Interphase correctly introduced as period between consecutive divisions; divided into sub-phases. **+0.5**

G1 phase: starts with production of new cell; cell grows in size; increases number of organelles; forms needed proteins and other substances; prepares for S phase (forms enzymes and nucleotides for DNA replication). **+1**

S phase: synthesis of DNA; cell replicates entire genetic material to form two copies; chromosomes each have two chromatids attached to one centromere — called sister chromatids. **+1**

G2 phase: cell prepares for mitotic phase; forms spindle proteins; accumulates energy for cell division; more mitochondria formed. **+0.75**

G0 phase: cell may exit the cell cycle; G0 cells stop dividing; some cells (neurons) enter G0 permanently (never divide again); cells of liver and kidneys enter G0 temporarily and can re-enter when needed. **+0.75**

Mitotic (M) phase: cell divides into daughter cells through karyokinesis followed by cytokinesis. **+0.5**

All sub-phases with key events. G0 phase correctly explained with neuron (permanent) and liver/kidney (temporary) examples.

5 / 5

Q.3(ii) — Main question

Karyokinesis in mitosis: all four stages with centromere role

4 / 5

Karyokinesis defined as division of nucleus forming two nuclei, divided into four stages (prophase, metaphase, anaphase, telophase). **+0.5**

Prophase (longest phase): hereditary material found as thin chromatin threads; during S phase, chromatin fibres duplicate; prophase chromatin fibres coil and condense into chromosomes; each chromosome has two chromatids attached at centromere; nucleolus disappears as DNA is packed into chromosomes; at end of prophase, nuclear membrane splits into vesicles which disperse in cytoplasm; spindle apparatus forms. **+1.5**

Metaphase: chromosomes aligned to equator of spindles; two spindle fibres, one from each pole, attached to centromere of chromosomes. **+0.5**

Anaphase: spindle fibres pull centromeres (which split); separated chromatids pulled from each other; individual chromatids pulled to respective poles. **+0.5**

Telophase: chromatids reach poles; they uncoil and lengthen to form chromatin fibres; spindle apparatus disintegrates; nuclear membrane reforms around chromatin at each pole; nucleoli reappear in both nuclei. **+0.5**

Cytokinesis: division of cytoplasm forming two daughter cells; begins while telophase is in progress; pattern differs in animal and plant cells. **+0.5**

Deduction: The role of the centromere as a mark-carrying structural point is not separately articulated. The answer describes centromere attachment of spindle fibres and chromatid attachment but does not explicitly state the centromere's role as the split point during anaphase (centromere splits, allowing sister chromatids to separate). The answer key awards 0.5 for an explicit statement of centromere function. -1

Q.3(ii) — Prophase detail is excellent. All four stages present. Deduction: centromere role not explicitly articulated as a separate functional statement (-1). **4 / 5**

Q.3(iii) — Main question

Meiosis I in detail: Interphase I, Karyokinesis I (all four stages), synapsis, crossing over, Cytokinesis I

5 / 5

Meiosis introduced as a continuous process for convenience divided into: interphase I, meiosis I, interphase II, meiosis II. **+0.5 (implicit in structure)**

Interphase I: before division, cells form a copy of their genome by replication; that is why chromosomes appear with two chromatids during division. **+0.5**

Meiosis I as reduction division: reduces chromosome number in daughter cells to half; karyokinesis forms two haploid nuclei followed by cytokinesis. **+0.5**

Karyokinesis I stages listed: Prophase I, Metaphase I, Anaphase I, Telophase I. **+0.5**

Prophase I (longest phase of meiosis I): chromosomes begin to appear as thicker fibres due to coiling of chromatin; homologous chromosomes line up point to point — centromeres in same position — this pairing is synapsis; each pair is a bivalent; non-sister chromatids join each other at points called chiasmata; each chiasma is a site for exchange of segments of non-sister chromatids — this is crossing over; by end of prophase I, chromosomes fully condensed; paired chromosomes repel but remain attached at chiasmata; nucleoli and nuclear envelope disappear; spindle apparatus forms. **+1.5**

Metaphase I: bivalents arranged at equatorial plate; spindle fibres attached to centromere of chromosomes; one member of a homologous pair receives spindle fibre from one pole, other from opposite pole. **+0.5**

Anaphase I: spindle fibres pull homologous chromosomes; separated chromosomes of each pair move to opposite poles; each pole receives haploid set. **+0.5**

Telophase I: at pole, chromosomes uncoil into chromatin; nuclear envelope reforms at each pole around haploid set. **+0.5**

Cytokinesis I: cleavage in animal cell; cell wall in plant cell; each daughter cell is haploid; chromosomes composed of two chromatids — not genetically identical due to crossing over; must be separated in meiosis II. **+0.5**

Outstanding coverage. Synapsis, bivalent, chiasma, crossing over all explicitly named and located. Prophase I is the standout section. Full marks.

5 / 5

Q.3(iv) — Main question

Compare and contrast mitosis and meiosis (6+ differences)

5 / 5

Two-column tabular format comparing mitosis and meiosis under the following headings:

(1) **Location**: somatic cells vs germ line cells. **+1**

(2) **Division**: cell divides only once vs cell divides twice. **+1**

(3) **Daughter cells**: produces 2 daughter cells vs 4 daughter cells. **+1**

(4) **Haploid/Diploid**: daughter cells are diploid vs daughter cells are haploid. **+1**

(5) **Crossing over**: no crossing over during mitosis process vs crossing over takes place. **+1**

(6) **Chiasmata formation**: no chiasmata formed vs chiasmata are formed. **(bonus / already covered under crossing over but shows depth)**

Six valid, clearly distinguishable criteria present using tabular layout. All entries accurate.

Six or more valid differences in clean tabular format. Full marks.

5 / 5

Strengths in this attempt

- **Perfect MCQ score (12/12)** — all 12 MCQs correct; no hesitation on any conceptual question including the bivalent composition (Q.7), the site of crossing over (Q.8), and the halving mechanism in meiosis (Q.10).
- **G0 phase correctly explained with two categories of cells** — neurons (permanent) vs liver/kidney (temporary), which many students omit entirely.
- **Prophase I is detailed and precise** — synopsis, bivalent, chiasmata, crossing over all explicitly named, placed in the correct stage, and causally linked. This is the hardest sub-topic of the chapter and was answered with full textbook precision.
- **Cytokinesis I: animal vs plant cell correctly distinguished** — cleavage furrow for animal, cell wall for plant, plus the important observation that daughter cells are not genetically identical and must proceed to meiosis II.
- **Clean two-column table layouts** — Q.2(xi) and Q.3(iv) both use well-structured comparison tables with accurate and distinct criteria.
- **Meiosis and chromosome number maintenance (Q.2(x))** — 46 chromosomes, meiosis produces 23, fertilisation restores 46; all figures correct.

Areas to improve

- **Q.3(ii): Centromere role not separately articulated** — the answer correctly places spindle fibre attachment at the centromere, but never writes a standalone sentence such as: "At anaphase, the centromere splits, allowing the two sister chromatids to separate as individual chromosomes." Explicit functional statements about the centromere carry their own mark in Section C. Always end the karyokinesis answer with a summary sentence on centromere function.
- **Q.2(viii)(or): Interphase II entry point slightly incomplete** — the fact that chromosomes in the daughter cells are NOT genetically identical was stated, but the explicit consequence (this is why meiosis II is necessary to separate them) was not fully connected. In future, write the link: "These two haploid daughter cells now enter interphase II. No further DNA replication occurs. Meiosis II separates the non-identical sister chromatids."

Comparison with Prior Biology Attempts

Attempt	Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
Bio-01	3 Apr 2026	exam-01-ch1-6	11/12	28/33	13/20	52/65	80.0%	A
Bio-02 (sit 1)	30 Apr 2026	exam-02-ch7	12/12	30/33	18.5/20	60.5/65	93.1%	A
Bio-02 (sit 2)	2 May 2026	exam-02-ch7	12/12	33/33	20/20	65/65	100%	A+
Bio-03 (this)	8 Jun 2026	exam-03-ch4	12/12	31.5/33	19/20	62.5/65	96.2%	A+

Trajectory note: Seemab's Biology scores show a clear positive trend: 80.0% (Ch1-6) to 93.1%/100% (Ch7, two sits) to 96.2% (Ch4, single attempt). This is a first-attempt score of 96.2% on a chapter (Cell Cycle) that is widely considered the most technical in Class 9 Biology — covering four stages of mitosis, all five stages of meiosis I in detail, G0 phase, and six criteria of mitosis vs meiosis comparison. The only marks dropped (2.5/65) relate to fine structural detail (centromere functional statement and the interphase II consequence sentence), not content gaps.